

Michael Serrano

michaelyserrano@gmail.com | (314) 471-8723 | Cambridge, MA | linkedin.com/in/michaelyserrano | github.com/michaelyserrano

Education

Massachusetts Institute of Technology, Cambridge, MA	GPA: 4.6/5.0
Master of Engineering Computer Science, Economics, and Data Science	January 2027
Bachelor of Science Physics, Computer Science, Economics, and Data Science	June 2026

Technical Skills

Languages: Python, Go, TypeScript, Rust, R, SQL, C

System Design: Dual-database architecture, async job queues, multi-stage data pipelines, self-healing execution loops, sliding-window rate limiting, graceful shutdown

Tools & Infra: PostgreSQL, Redis, Docker, GitHub Actions, Railway, Cloudflare, Sentry, Playwright, AWS, Celery, Airflow

Protocols & Auth: REST APIs, MCP, OAuth 2.0 PKCE, JWT, AES-256-GCM encryption

ML/AI: PyTorch, TensorFlow, scikit-learn, LangChain, Claude API, Snorkel, NumPy, Pandas, PySpark

Work and Research Experience

Co-Founder, Software Engineer | Rindler (YC S26) (*San Francisco, CA*) November 2025 — Present

- Built an agent-agnostic MCP server in Go with OAuth 2.0 PKCE, JWT auth, and AES-256-GCM encryption at rest — any AI model can search, navigate, and transact on retail sites without per-site integration
- Deployed a Playwright worker fleet with Redis/async priority queues, ephemeral browser containers, and a self-healing execution loop with automatic fallback promotion
- Architected dual-database separation (control plane vs. data plane) with structured logging (slog), Sentry error tracking, and sliding-window rate limiting
- Built GitHub Actions CI/CD gating PRs before Railway auto-deploy, with Cloudflare Pages/Workers at the edge
- Achieved **4x faster** (407s→98s) and **6x cheaper** (\$1.54→\$0.26) task completion vs. open-source agents in benchmark arena

Machine Learning Intern | Roblox (*San Mateo, CA*) May — August 2025

- Developed LSH-minhash + logistic regression pipeline detecting alt accounts across **150B user-pairs**
- Introduced Snorkel-based weak supervision for training data, cutting model release cycle from **months to 1 hour**
- Improved alt-account detection recall from **94% to 96%** at 90% precision via refined candidate generation
- Rebuilt and documented defunct training pipeline, enabling rapid PR curve-evaluated retraining

LLM Researcher | MIT CSAIL (*Cambridge, MA*) January 2025 — Present

- Built and deployed ConceptQ to MIT course staff with map-reduce concept graph extraction, interactive canvas editing, and agentic Bloom's Taxonomy-aligned question generation via RAG
- Designed async infrastructure (Celery + Redis Streams) with live SSE streaming of generated questions
- Integrated NextAuth v5 with CSAIL OIDC and MIT Touchstone for production auth with three-tier RBAC
- Engineered automated prompt optimization loop evaluated on a 6-dimensional rubric for iterative prompt tuning

Machine Learning Intern | Better Off Social (*New York, NY*) June — August 2024

- Trained a GCN to partition event attendees into compatible groups; **99.2%** satisfaction rating in production
- Developed a PyTorch feedforward network to predict pairwise compatibility from demographic and personality features
- Refactored matching algorithm codebase and documentation, yielding a **54x speed-up**

Algorithmic Game Theory Researcher | MIT Sloan (*Cambridge, MA*) January — May 2023

- Profiled Deep Q-Network dynamic pricing agents and characterized emergent collusive pricing behavior
- Analyzed how varying levels of prior market knowledge shift Nash equilibrium outcomes between competing agents

Leadership and Extracurriculars

Vice President and Treasurer | Sloan Business Club (*Cambridge, MA*) January 2024 — December 2025

- Managed club expenditures and a \$20,000 budget for events, marketing, and recruitment
- Led outreach for external speakers and logistics for meetings and conferences

VP of Finance, Risk Management Director | Beta Theta Pi (*Cambridge, MA*) January 2023 — December 2025

- Managed a **\$400,000** budget for house upkeep, brotherhood initiatives, and asset management
- Designed and launched a flex-spending allocation system for house improvements, increasing member retention